TEXORA · LECTURE SHEET

Jute Carding

Carding turns the softened, entangled jute reed into a spinnable sliver. Because the reed is metres long, stiff and gummed, jute uses great pinned rollers in two stages — breaker and finisher — where cotton uses a single wire-clothed card.

TE-2101 · SPINNING TECHNOLOGY-I · TOPIC 6 · LEVEL 2 TERM 1

1. Why jute carding is not cotton carding

Carding has the same objects in both fibres — open the material, remove what should not go forward, parallelise the fibres and deliver a uniform sliver — but the material dictates wholly different machines. Five differences decide the design:

- **Fibre-unit length.** Cotton arrives as individual fibres 20–40 mm long; a card working point only ever handles a tuft of such fibres. Jute arrives as reeds 1.5–3.5 m long — longer than the whole circumference of a card cylinder. The feed must therefore present the reed *across* a lattice in controlled lengths, and opening must tear a network apart progressively, not comb a tuft.
- **Stiffness and cement.** Even after softening (Topic 5), the reed network resists bending and separation. Fine saw-tooth wire clothing, which works on cotton by point density and close settings, would choke, strip its teeth and be destroyed by jute's woody defects. Jute cards use **stout pins set in wooden lags or rollers** — pinned rollers strong enough to grip, tear and comb reeds, and cheap to re-pin when damaged.
- **Entanglement and defects.** The feed is a matted sheet containing sticks, hurd and root pieces. The first card must be an aggressive *breaker*: its job is as much demolition and waste extraction as fibre alignment.
- **One card is not enough.** A single passage cannot both break down metre-long reeds and deliver a fine, even sliver. Jute carding is therefore in **two stages**: the **breaker card** opens and cleans, the **finisher card** — fed with breaker sliver — combs finer and evens the product.
- **Sliver scale.** Jute slivers are heavy beyond cotton experience: hundreds of grams per metre, against a few grams per metre for cotton. Every production and draft calculation in the jute card room runs an order of magnitude larger.

2. The breaker card — construction

The breaker card is a large roller card. Following the material from feed to delivery:

- **Feed lattice and dole.** The softened sheet of reeds from the spreader is carried forward on an endless lattice (the **dole** arrangement presents weighed portions of reed in a steady stream). The lattice meters the feed: its speed, with the weight laid on it, fixes the input rate and hence the sliver weight.
- **Feed rollers.** A pair of heavy, often pinned or fluted rollers grips the sheet at the lattice end and presents it slowly to the cylinder, holding back the mass while the pins take the leading ends — the grip-and-tear point where the reed network first begins to open.
- **Cylinder.** The great pinned roller at the machine's heart, roughly a metre or more in diameter, clothed with pins set in lags. It carries fibre forward under the worker rollers at high surface speed and does most of the opening work.
- **Worker and stripper pairs.** Around the top of the cylinder sit several pairs of smaller pinned rollers. Each **worker** turns slowly against the cylinder's motion: at the worker–cylinder nip the strong relative surface-speed difference combs, splits and opens the fibre — the true carding action of the machine. The worker retains fibre on its pins; its partner, the **stripper**, turns fast, strips the worker clean, and returns the fibre to the cylinder to be carried to the next pair. Successive pairs repeat the action until the fibre reaches the doffer in a far more open, parallel state than it entered.
- **Doffer.** A further pinned roller that takes the opened fibre off the cylinder (condensing it from the cylinder's wide, fast surface onto its own slower one) and delivers it as a web.
- **Delivery.** The web is stripped from the doffer, condensed through a trumpet or between delivery rollers, and formed into the heavy breaker **sliver**, coiled or laid into cans for the finisher card.

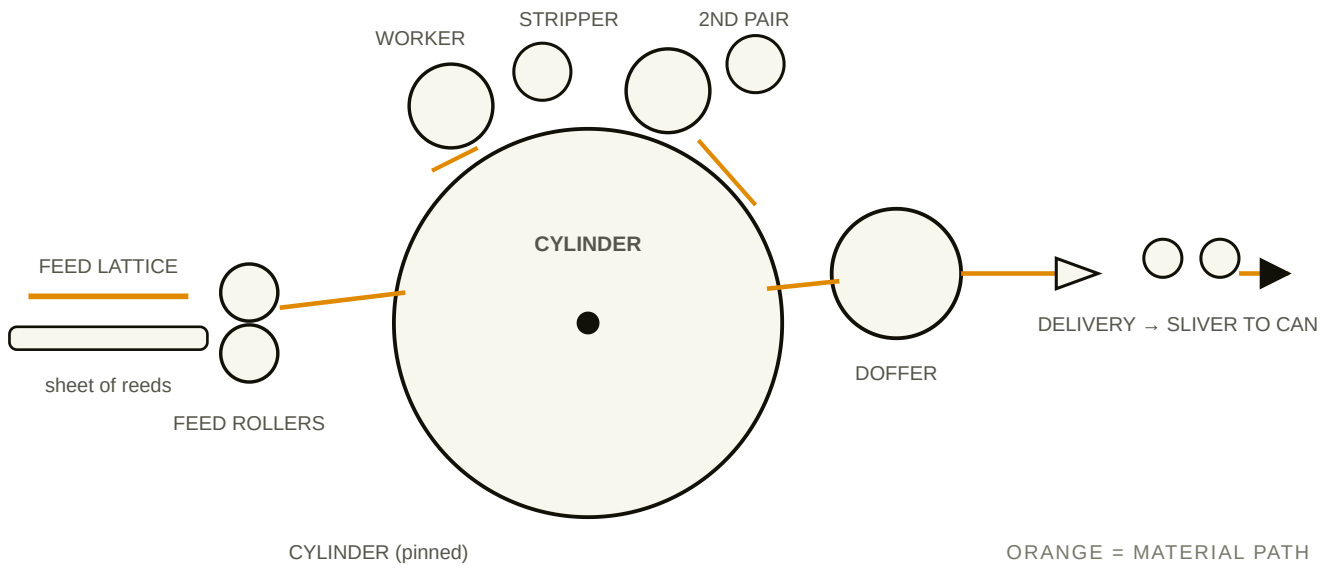


Figure 1 — Passage of material through a jute breaker card, cross-section. Workers comb against the cylinder; strippers clean the workers and return the fibre; the doffer condenses the opened fibre into the delivery web.

3. The breaker card at work

At the feed rollers the cylinder's pins seize the projecting ends of the gripped sheet and **tear tufts away** against the feed rollers' hold — the first, coarsest opening, in which long reeds are drawn out of the mat and partly split lengthwise. Carried under the first worker, the fibre meets a surface moving slowly against it: pins interpenetrate, bundles are combed apart, the reed's cemented cells begin to separate into finer strands, and sticks and loose dirt are thrown or fall as waste beneath the machine. The stripper clears the worker and hands the fibre back to the cylinder; the second and third worker pairs repeat the combing on an already finer material. By the doffer, the metre-long reed network has become a lofty web of much finer, roughly parallel fibre strands. Two cautions govern the whole action: the softening of Topic 5 must have done its work, or this tearing is breakage; and the feed must be even, because every thick place on the lattice becomes a thick place in the sliver that no later doubling wholly removes.

4. The finisher card

The finisher card is built on the same plan — lattice feed, feed rollers, pinned cylinder, worker–stripper pairs, doffer, sliver delivery — but everything is set for refinement rather than demolition:

- **Feed** is not raw reed but breaker sliver, delivered from cans (several ends are commonly creelled together, giving a first doubling that already averages breaker irregularities).
- **Pinning is finer and denser** — smaller pins, closer set in the lags — because the material is already opened strands, not reeds; fine pins comb the strands toward ultimate fibres without the violence the breaker needed.
- **Settings are closer and speeds relatively higher** at the working points, so the carding action per unit of fibre is more intensive.
- **Delivery** is a cleaner, finer, more uniform sliver, passed to the first drawing frame. Finisher sliver weight is lower than breaker sliver weight, and its evenness — not its cleanliness alone — is what the drawing passages and the spinning frame will build on.

Not every mill flow uses both cards at full duty for every product: the coarsest sacking work may take breaker sliver more directly, while fine work guards the finisher passage jealously. But the two-stage principle — break first, finish second — is the standard answer to jute's length and stiffness.

5. Pins, densities and speeds — the principles

Three working variables, set together, decide how a jute card behaves:

- **Pin density and size.** Pins per unit area of roller surface, and their length, thickness and angle. The breaker uses **coarser, sparser, stronger pins**: they must survive sticks and root pieces and grip whole reeds. The finisher uses **finer, denser pins** for combing opened strands. Within either card,

pins generally become finer from feed to delivery, matching the material as it opens. Worn, bent or missing pins are not a cosmetic fault: each dead patch passes an unopened bundle forward as a sliver defect.

- **Roller surface speeds.** The cylinder runs fastest and carries the fibre; workers run **much slower** in surface speed, so the speed *difference* at the cylinder–worker nip creates the combing draft that opens the fibre; strippers run fast enough to clear their workers completely; the doffer runs slower than the cylinder so fibre condenses onto it, and the delivery runs slower still, setting — with the feed rate — the sliver weight. Raising cylinder speed raises production but shortens the working time at each pair and increases fibre breakage and fly; the worker speed chiefly controls intensity of carding.
- **Settings (roller distances).** The gaps between feed rollers and cylinder, workers and cylinder, and doffer and cylinder are set in small fractions that decide how deeply the pin fields interpenetrate. Closer settings card more thoroughly — and break more fibre and wear more pins. Settings are therefore chosen with the batch: a strong, well-softened batch tolerates close work that would destroy a weak, harsh one.

6. Card waste

Jute carding throws out a substantial waste, most of it at the breaker card. Its components are collectively part of the mill's **caddis** — the sweepings and droppings of the spinning preparation: short fibre torn out in opening, sticks and hurd shaken from the reeds, root fragments, dust and dirt, and fibre fly collected around the rollers. Some caddis is not waste at all in the economic sense: sound short fibre from it is re-used in the coarsest batches (for example for low-grade sacking weft or backing yarns), while the truly woody and dirty fraction is discarded or sold for non-spinning uses. Waste percentage is a quality instrument, not just a loss figure: a breaker card producing too little waste is passing sticks and short fibre into the sliver; one producing too much is breaking good fibre — usually a softening fault (Topic 5), blunt or damaged pins, or settings too close for the batch. The card tenter's routine — clearing droppings, checking pin condition, weighing waste against feed — is how that balance is held shift by shift.

7. Sliver weights and production

Jute card sliver is weighed in **grams per metre** and runs to **hundreds of g/m** — breaker deliveries commonly of the order of 200–400 g/m and finisher deliveries somewhat lower, against roughly 3–6 g/m for a cotton card sliver. Two consequences follow. First, draft at the jute card (feed weight per metre ÷ delivery weight per metre, over the machine's surface-speed ratios) is modest compared with the opening work done; the card's main product is *opened fibre*, not attenuation. Second, production is large in mass terms and follows directly from delivery speed and sliver weight:

Production (kg/h) = delivery speed (m/min) × 60 × sliver weight (g/m) ÷ 1000

WORKED EXAMPLE — FINISHER CARD PRODUCTION

A finisher card delivers sliver of 220 g/m at a delivery speed of 55 m/min.
Length delivered per hour = 55 m/min × 60 min = 3,300 m.
Mass = 3,300 m × 220 g/m = 726,000 g.
Production = 726,000 ÷ 1000 = **726 kg/h** — i.e. 55 × 60 × 220 ÷ 1000 = 726 kg/h.
Second check (breaker card): sliver 300 g/m at 45 m/min → length per hour = 45 × 60 = 2,700 m; mass = 2,700 × 300 = 810,000 g = **810 kg/h** (45 × 60 × 300 ÷ 1000 = 810 kg/h). The breaker out-produces the finisher in mass because its sliver is heavier — the finisher's lighter delivery, not a slower machine, accounts for the difference.

8. Cotton card and jute cards compared

FEATURE	COTTON CARD	JUTE BREAKER CARD	JUTE FINISHER CARD
Feed material	Lap of opened tufts; single fibres 20–40 mm	Sheet of softened reeds 1.5–3.5 m long	Breaker sliver from cans (doubled ends)
Clothing	Flexible / saw-tooth wire clothing, very fine points	Stout pins in lags/rollers, coarse and sparse	Pins, finer and denser than breaker
Main working elements	Licker-in, cylinder with flats (or revolving flats)	Pinned cylinder with worker–stripper roller pairs	Pinned cylinder with worker–stripper pairs, closer settings
Main action	Individualising single fibres; nep and trash removal	Tearing the reed network open; removing sticks and roots	Combing strands toward ultimate fibres; evening the sliver
Waste	A few % (flat strips, droppings)	High — caddis: short fibre, sticks, root, dust	Low — small droppings and fly
Delivered sliver	≈ 3–6 g/m	Hundreds of g/m (≈ 200–400 g/m)	Hundreds of g/m, lower than breaker
Stages for spinning	One card	Two cards in sequence — break, then finish	

Read the table as a lesson in material-led machine design: every row differs because the feed differs. Put cotton through a jute card and its short fibres would be lost in the pinning and waste; put jute through a cotton card and the wire clothing would be wrecked by the first bale of reeds.

Glossary

Breaker card — the first jute card: coarse pinned rollers that tear the softened reed network open and deliver heavy sliver. **Finisher card** — the second card: finer, denser pinning that combs breaker sliver into a uniform delivery sliver. **Worker** — the slow pinned roller that combs against the cylinder. **Stripper** — the fast roller that cleans its worker and returns fibre to the cylinder. **Doffer** — the roller that condenses opened fibre off the cylinder into a web. **Dole / feed lattice** — the lattice feed presenting weighed reeds to the feed rollers. **Caddis** — jute preparation waste and sweepings: short fibre, sticks, root and dust. **Pin density** — pins per unit area of roller surface; coarser at the breaker, finer at the finisher. **Sliver weight** — mass per unit length of sliver (g/m); hundreds of g/m in jute carding.

Key points

• Jute carding differs from cotton carding because the feed is a long, stiff, gummed reed — pinned rollers, not wire clothing, and two cards, not one.
• Breaker card: feed lattice/dole → feed rollers → pinned cylinder → worker–stripper pairs → doffer → heavy sliver.
• The carding action happens at the cylinder–worker nip: slow worker against fast cylinder combs and splits the fibre; the stripper keeps the worker clean.
• The finisher card repeats the plan with finer, denser pins and closer settings, fed from breaker sliver cans, to even the sliver for drawing.
• Pin density, roller surface speeds and settings are chosen together, and matched to the batch's strength and softening.
• Card waste (caddis) is a control figure: too little means dirt goes forward, too much means good fibre is being broken.
• Production (kg/h) = delivery (m/min) × 60 × sliver (g/m) ÷ 1000 — jute slivers of hundreds of g/m make card production hundreds of kg/h.

Practice

1. A breaker card delivers 350 g/m at 40 m/min. Calculate its production in kg/h, showing each step.
2. A finisher card must produce 600 kg/h of 200 g/m sliver. What delivery speed (m/min) is required?
3. Explain why a jute card uses pinned rollers rather than the saw-tooth wire clothing of a cotton card. Give two distinct reasons.
4. At one worker position the stripper is found to be running glazed with fibre. What happens to the carding action at that pair, and what defect appears in the sliver?

5. Breaker-card waste rises sharply on a new batch while settings are unchanged. List three batch-side or machine-side causes, and state which you would check first and why.
6. Why does doubling breaker sliver ends at the finisher feed improve evenness even before the finisher's own carding action is considered?

Answer key

1. Length per hour = 40 × 60 = 2,400 m. Mass = 2,400 × 350 = 840,000 g. Production = 840,000 ÷ 1000 = **840 kg/h** (40 × 60 × 350 ÷ 1000 = 840).
2. From production = speed × 60 × weight ÷ 1000: speed = 600 × 1000 ÷ (60 × 200) = 600,000 ÷ 12,000 = **50 m/min**.
3. (i) Strength against the feed: metre-long reeds with sticks and root pieces would choke and strip fine wire teeth; stout pins survive and can be re-pinned. (ii) Grip and tear: the reed network must be gripped and torn apart, which needs long, strong, widely spaced points; wire clothing is designed to individualise already-short single fibres, and jute's fibres-in-reed would load and roll on it instead of opening.
4. The stripper no longer clears the worker, so the worker loads up with fibre and stops combing — its pins no longer penetrate and open the material at the cylinder nip. Fibre passes that pair essentially uncarded and is returned unevenly, appearing in the sliver as unopened bundles / thick, streaky places (and rising fly and waste at that position).
5. Any three: batch under-softened or unevenly conditioned (harsh reeds break instead of opening); batch weaker or shorter than the standard the settings suit (more fibre torn out as caddis); pins damaged or worn so they break rather than comb; feed laid too thick so the card is overloaded. Check softening/conditioning first — it is the commonest cause, it affects the whole batch rather than one position, and Topic 5's faults present exactly as excess card waste and breakage.
6. Doubling averages independent irregularities: a thick place in one end is laid against the average of the others, so the combined feed's mass per metre varies less than any single end's. The finisher therefore starts from a more even input, and its delivery variation is reduced by the same averaging — the principle later exploited at every drawing passage.
